ISO 9241-210 Human-Centred Design Report



MANIPAL INSTITUTE OF TECHNOLOGY
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FreshTrack

Food Inventory and Expiry Manager

CSE 4453 : Human Computer Interface

	Student Name	**Roll No.**
1)	Abhiraj Sengupta	42
2)	Abhi Verma	56
3)	Akhand Pratap Singh	55
4)	Satvik Garg	50

Section F | VII Semester B.Tech CSE
School of Computer Engineering
Manipal Institute of Technology, Manipal

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1. Introduction

1.1 Project Overview

FreshTrack is a mobile interface concept that helps households and hostel residents keep track of the food they store, see what has to be used first, and plan shopping around what is already at home. The design covers adding groceries, an inventory sorted by expiry date, colour-and-icon priority indicators, a daily expiry reminder, recipe suggestions that use soon-to-expire ingredients, a generated shopping list, and the mark-as-used and remove-item interactions.

We produced sixteen high-fidelity screens and an interactive prototype that follows one sample household pantry through the complete journey. The report follows the ISO 9241-210 human-centred design report format [1] and applies the design rules, task-analysis, dialogue and evaluation techniques covered in the course [2].

1.2 Problem Statement

Groceries bought in a weekly or monthly trip are spread across a fridge, a freezer and a shelf. Perishable items such as milk, paneer, curd and leafy vegetables are pushed to the back and forgotten until they spoil, while items that are still at home are bought again because nobody remembers they are there. In hostels the fridge is shared, so no single person has the full picture. Household food waste in India has been estimated at about 50 kg per person per year [3].

Existing note-taking or generic list apps do not solve this, because they show items without urgency, require long data entry for every item, and do not connect what is stored with what to cook or what to buy. The problem addressed is therefore: **how can a person record food with minimal effort and be shown, at the right moment, which items need to be used first?**

1.3 Objectives

- Make it obvious, at a glance, which items are expired, need using now, need using soon, or are still fresh (visual prioritisation).
- Reduce data entry for adding an item to a single tap in the common case, with sensible defaults for location and use-by date.
- Prevent common mistakes: duplicate purchases, past or mistyped dates, and confusing “eaten” with “wasted”.
- Remind users once a day at a time they choose, grouped into a single notification instead of one alert per item.
- Connect the inventory to action: suggest recipes that use expiring food and generate a shopping list from what was used or wasted.
- Allow every removal to be reversed (undo) instead of asking “Are you sure?”.

1.4 Scope

The scope is the user-facing mobile flow for one household or shared hostel fridge. The deliverable is a high-fidelity prototype with sixteen screens, together with the supporting design artefacts (sketches, wireframes, information architecture, user flow, task analysis and a state-transition model) and an expert usability evaluation of the inventory workflow.

Notifications, barcode scanning and recipe data are represented in the prototype but are not connected to live services. Item names, dates and quantities are demonstration values for a sample date of Thursday, 30 October 2026. Multi-user sync, a mess/canteen dashboard, nutrition tracking and online grocery ordering are outside the current scope.

2. Understanding the Context of Use

2.1 Target Users

ISO 9241-11 defines usability as the extent to which specified users can achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use [4]. We therefore describe the users, environment and tasks first. Following the stakeholder categories discussed in the course [2], the users are grouped as follows.

Table 1. Stakeholders

Stakeholder group	Who	Relationship to FreshTrack
Primary users	Hostel residents and students living in shared flats; household members who buy and cook food	Add groceries, check what to use first, mark items as used, plan shopping
Secondary users	Roommates or family who share the same fridge	Benefit from the shared list and reminders; occasionally mark items as used
Tertiary stakeholders	Campus store, local grocers, hostel mess	Affected by changed buying patterns; not users of this prototype
Facilitating stakeholders	Design and development team; course evaluators	Design, build and evaluate the system

Users are mostly comfortable with smartphones but have very different levels of patience for data entry. Some users, such as parents managing a family kitchen, may be less experienced with apps, so the design cannot rely on learned gestures or hidden features. About 8% of men have some form of colour-vision deficiency, so urgency must never be shown by colour alone.

2.2 User Environment

Table 2. Context of use

Situation	Physical and social context	Design implication
Right after shopping	Standing at the fridge, bags in hand, several items to put away quickly	One-tap adding, defaults for location and expiry, stay on the add screen to add the next item
Evening, deciding what to cook	At home, around 6–9 pm, often tired; may be cooking with roommates	One grouped reminder at a chosen time; recipe ideas that use what is expiring
At the shop	Walking in an aisle, one hand free, intermittent attention	Large checkboxes, "already at home" warnings visible in the list
Quick check during the day	Between classes, glancing for a few seconds	Home screen summary readable in a glance; priority shown by colour, icon and words

The device is a smartphone used one-handed in portrait orientation (designed at 390 × 844 points). Interruptions are frequent, so every screen must show its own state and nothing important may depend on remembering a previous screen.

2.3 User Tasks

We analysed the goal "keep household food from being wasted" using Hierarchical Task Analysis (HTA) [2]. The goal breaks down into five sub-tasks, each with its own plan (Figure 1). The analysis showed that task 1 (recording groceries) is performed most often and with the most items at a time, so it gave the strongest reason to minimise data entry. Tasks 2–4 happen daily and are triggered by time, which is why a reminder and an urgency-sorted view are central to the design.

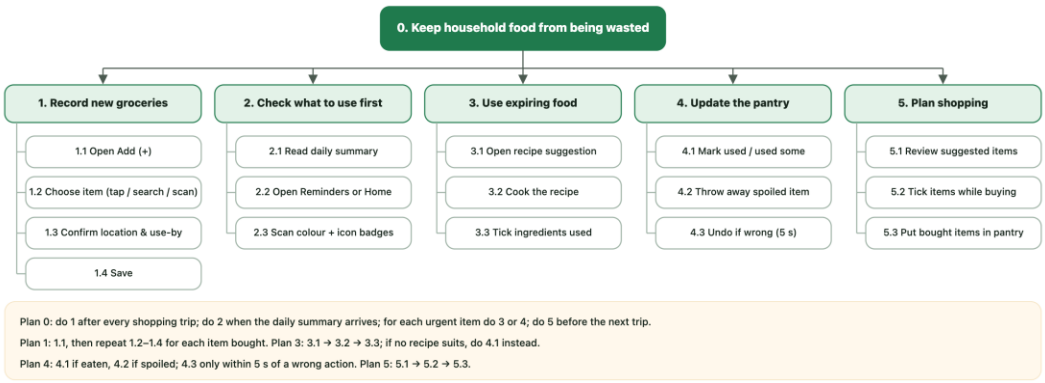


Figure 1. Hierarchical task analysis of the main goal, with plans.

Table 3. User tasks

ID	User task	Frequency
T1	Add newly bought items with location and use-by date	After every shopping trip (5–15 items)
T2	Find out which items must be used first	Daily
T3	Mark an item as used, partly used or thrown away; correct a mistake	Several times a day
T4	Choose a meal that uses expiring ingredients	Most evenings
T5	Set when and how often to be reminded	Once, at first use
T6	Build a shopping list and avoid buying what is already at home	Before every shopping trip
T7	Move bought items from the list into the pantry	After every shopping trip

3. User Research

3.1 Personas

This project uses two proto-personas and a scenario-based empathy map, built from the problem brief and the team's own experience of hostel and household food storage. No formal interviews, surveys or observation studies were conducted; the personas are assumptions to be validated in the usability test described in Section 7.2.

Table 4. Personas

Persona	Scenario and goal	Design response
Persona A — Riya, 20 B.Tech student, hostel resident; shares a fridge with three roommates	Buys milk, paneer, bread and fruit from the campus store twice a week. Often finds spoiled paneer behind someone else's food and has bought curd that was already in the fridge. Wants to use what she has before it spoils without "managing" an app.	One-tap quick add with default expiry; Home shows "Use these first"; duplicate warning when adding; shopping list warns about items at home.
Persona B — Prakash, 46 Manages groceries for a family of four in Manipal; occasional app user	Buys in bulk once a month and keeps a lot in the pantry and freezer. Turned off a previous app because it sent a notification for every item. Wants one clear daily reminder and large, readable controls.	One grouped evening summary at a chosen time; large touch targets; plain words such as "Used it all" and "Threw it away".

3.2 Empathy Map

The empathy map focuses on Persona A, because the hostel context combines a shared fridge, small quantities and irregular routines.

Table 5. Empathy map for Persona A

Dimension	Assumed perspective	Design implication
Says	"Whose paneer is this? It's gone bad again." "I didn't know we still had curd."	Show quantity and location per item; warn before adding a duplicate.
Thinks	"Typing every item is not worth the effort." "I'll remember to use the milk tomorrow."	Make adding faster than the effort it saves; do not rely on memory — remind at a fixed time.
Does	Puts groceries away in a hurry; checks the fridge only when hungry; shops without a list.	Quick add stays open for the next item; Home answers "what should I eat first?"; list generated automatically.
Feels	Guilty about wasting food and money; annoyed by nagging notifications.	Positive feedback ("90% eaten in time"); one summary instead of many alerts; waste logged neutrally.

4. User Requirements

4.1 Functional Requirements

Table 6. Functional requirements

ID	Requirement represented in the design	Screen
FR1	Add an item in one tap from frequent items, by search with autocomplete, or by barcode; location and use-by date are pre-filled from typical shelf life and can be changed.	S5
FR2	Warn when the item being added is already in the pantry, and show its remaining time.	S6
FR3	Detect a use-by date that is already in the past and offer a one-tap correction.	S7
FR4	Show the inventory sorted by expiry, grouped into Expired / Use now / Use soon / Fresh, filterable by location.	S4
FR5	Summarise the pantry by priority on Home and list the items to use first.	S3
FR6	Mark an item as used, partly used or thrown away, or remove it if it was added by mistake; offer Undo after every removal.	S8, S9
FR7	Send one grouped daily reminder at a user-chosen time, with a chosen warning period (1–3 days).	S2, S10, S11
FR8	Suggest recipes ranked by how many soon-to-expire items they use; never suggest expired food.	S12, S13
FR9	After cooking, update all used ingredients in one step.	S14
FR10	Generate shopping suggestions from used, wasted and missing items; warn about items already at home; move bought items to the pantry with usual expiry dates.	S15, S16

4.2 Non-Functional Requirements

Table 7. Non-functional requirements

Quality	Requirement	How it is measured
Efficiency	Adding a common item takes no more than 6 s; marking an item as used takes one tap from Home or Pantry.	KLM prediction (Section 7.1) and timed test
Learnability	A first-time user can add five items and identify the most urgent item without help.	Task success in usability test
Error tolerance	Every removal can be undone for 5 s; past dates and duplicates are caught before saving.	Walkthrough; error count in test
Accessibility	Priority is coded by colour, icon shape and words; text contrast is at least 4.5:1 (WCAG AA); touch targets are at least 44 pt; animation respects the reduced-motion setting.	Contrast check; greyscale check (Figure 7)
Consistency	The same four priority levels, badges and wording are used on every screen and in notifications.	Heuristic review
Privacy	Pantry data stays on the device in this version; sharing the list is an explicit user action.	Design review

4.3 Prioritization

Requirements were prioritised with the MoSCoW method against the two personas and the HCI focus areas in the brief.

Table 8. Prioritisation (MoSCoW)

Priority	Scope
Must have	Quick add with defaults (FR1), priority-sorted inventory (FR4), Home summary (FR5), mark-as-used / remove with undo (FR6), daily reminder (FR7).
Should have	Duplicate and past-date checks (FR2, FR3), shopping list with “already at home” warning (FR10), recipe suggestions (FR8).
Could have	Barcode scanning, “I cooked this” bulk update (FR9), monthly used/wasted feedback, sharing the list with roommates.
Won’t have (now)	Multi-device sync, nutrition tracking, online grocery ordering, mess/canteen dashboard.

5. Design Solutions

5.1 Information Architecture

The interface is organised around the questions a user asks, not around data types: “What should I use first?” (Home), “What do I have?” (Pantry), “What did I just buy?” (Add), “What can I cook?” (Recipes) and “What do I need?” (Shopping). These five areas form the tab bar, with Add as the central action. Reminders open from the bell on Home and from the daily notification. A two-step first-run flow sets the reminder preferences once (Figure 2).

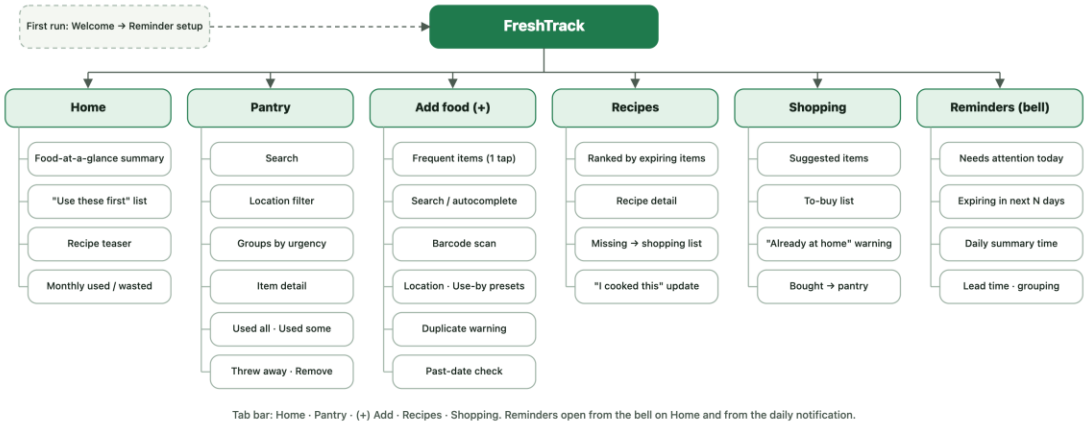


Figure 2. Information architecture of FreshTrack.

5.2 User Flow

Figure 3 shows the three main flows: adding groceries, using food before it expires, and planning shopping. Decision points where a mistake is likely (duplicate item, past date, item already at home) are handled in the flow with a warning rather than a blocking error. The flows meet at the same goal state: a pantry sorted by expiry.

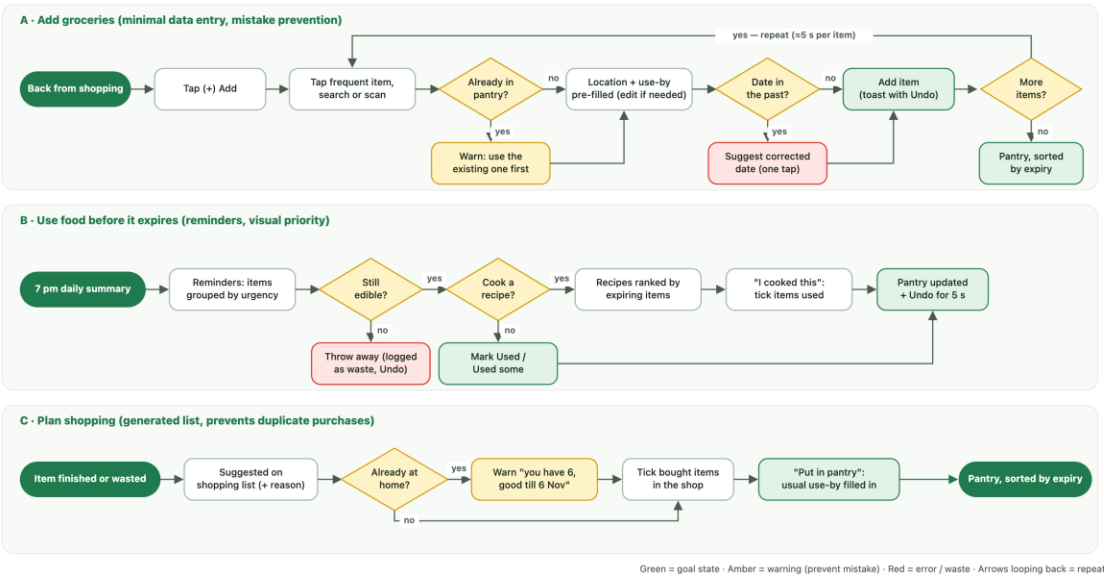


Figure 3. User flows for adding groceries, using food and planning shopping.

To check the dialogue for unreachable or dead-end states, we modelled the life of a single food item as a state transition network (STN) [2] (Figure 4).

Time moves an item from Fresh to Use soon, Use now and Expired. User actions move it to Used, Wasted or Removed. Every removal has an Undo path back to the previous state, and Used and Wasted items can go to the shopping list and back into the pantry. The model confirmed that no state is a dead end: even an expired item has two exits.

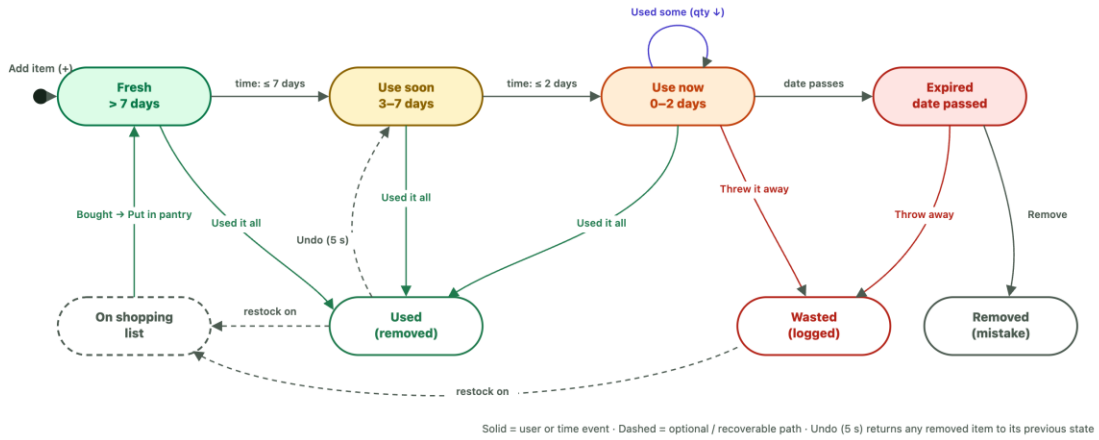


Figure 4. State transition network of a food item.

5.3 Low-Fidelity Sketches

Paper-style sketches explored the four key screens before any visual design (Figure 5). They fixed the main ideas: an urgency summary at the top of Home, a grid of frequent items for one-tap adding, an item screen that asks “what happened to it?” instead of offering only Delete, and a shopping list that knows what is already at home.

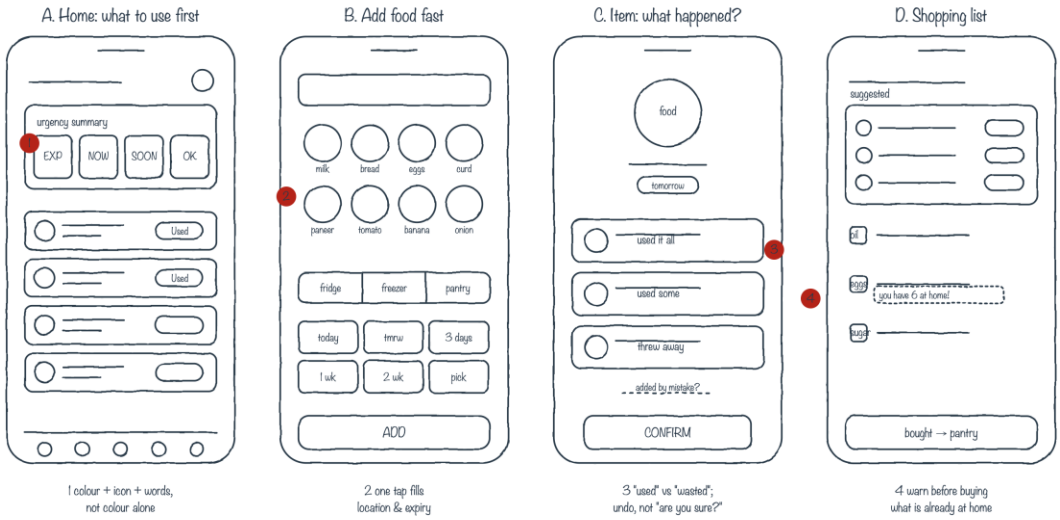


Figure 5. Low-fidelity sketches of Home, Add food, Item and Shopping list.

5.4 Wireframes

A first set of greyscale wireframes (v1) followed a conventional "inventory app" pattern: an alphabetical list, a form with six required fields, a Delete button with a confirmation dialog, one notification per item and a manual shopping list (Figure 6). We inspected these wireframes with the techniques in Section 7. The seven numbered problems became the main design changes and are listed in Section 7.3.

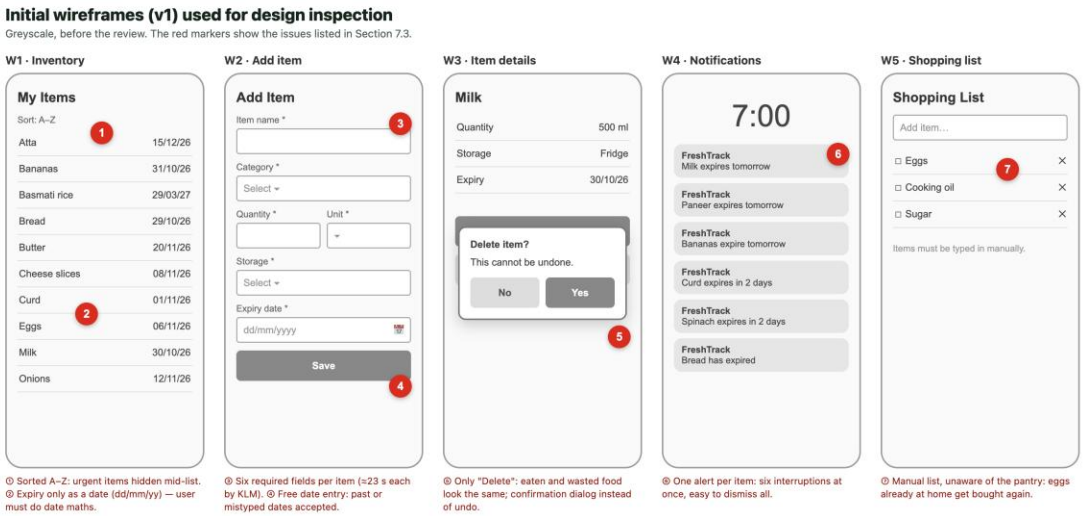


Figure 6. Initial wireframes (v1) used for design inspection.

5.5 High-Fidelity Prototype

The final design uses a light green-grey background, dark green text and a deep green primary action. Urgency is the only strong colour on the screen: four priority levels, each with its own colour, icon shape and wording (Figure 7). In greyscale the levels can still be told apart by icon and words, which supports colour-blind users and black-and-white printing. All badge text meets WCAG AA contrast (4.5:1 or higher). Lists are always sorted soonest-first, and each row repeats the same structure: food icon, name, quantity and location, priority badge, and a one-tap "Used" action.

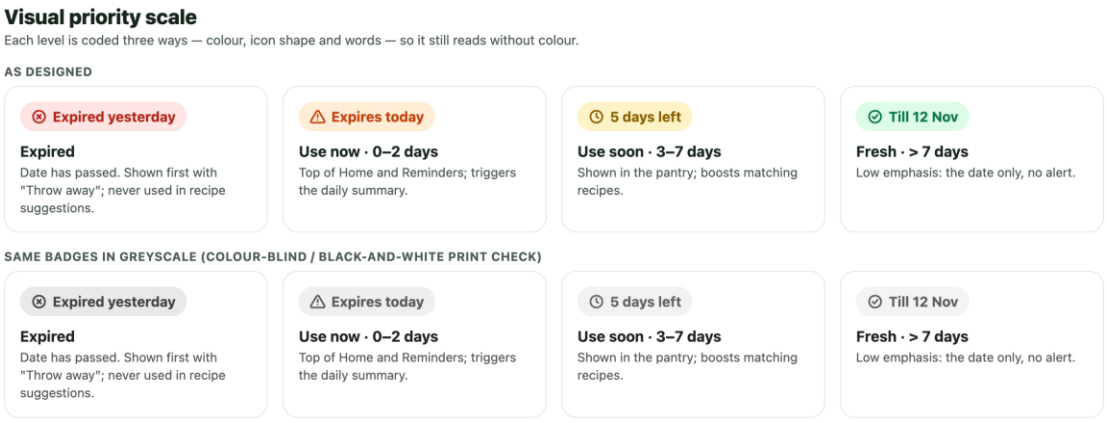


Figure 7. Visual priority scale, in colour and in greyscale.

Single-choice settings with few options (storage location, how much is left, reminder time and warning period) use a segmented chip control (JellyRadio from the React Bits library) with large 44-pt touch targets. The selected option swells slightly, which gives immediate feedback. Two decorative animated components appear only on the first-run screens: an animated food spiral inside pulsing rings on Welcome (InfiniteSpiral, MagicRings), and a field of lines that follow the finger on Reminder setup (MagnetLines). They are kept off the working screens so motion never competes with the urgency colours, and all of them stop when the phone's reduce-motion setting is on.

Table 9. Sample household used across all screens

Sample data	Illustrative value
Date	Thursday, 30 October 2026, 6:45 pm; daily summary at 7:00 pm
Pantry	16 items: 1 expired (bread), 5 use now (milk, paneer, bananas, curd, spinach), 2 use soon (tomatoes, eggs), 8 fresh
Priority rules	Expired: date passed · Use now: 0–2 days · Use soon: 3–7 days · Fresh: more than 7 days
Recipes	Palak Paneer uses 3 expiring items (ranked first); Banana Milkshake uses 2; expired bread is never suggested
Shopping	Suggested: bread (thrown away 29 Oct), milk (runs out today), fresh cream (missing for Palak Paneer); eggs flagged as "6 at home"

The sixteen final screens are shown in Figures 8–23. They follow one sample household from first launch to restocking.

Final screens 1 and 2

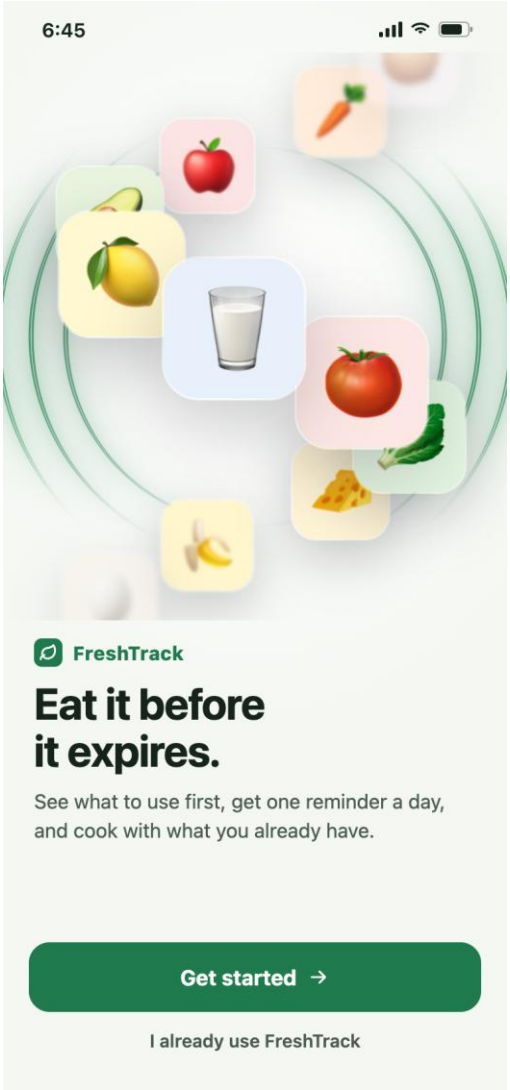


Figure 8. S1 Welcome

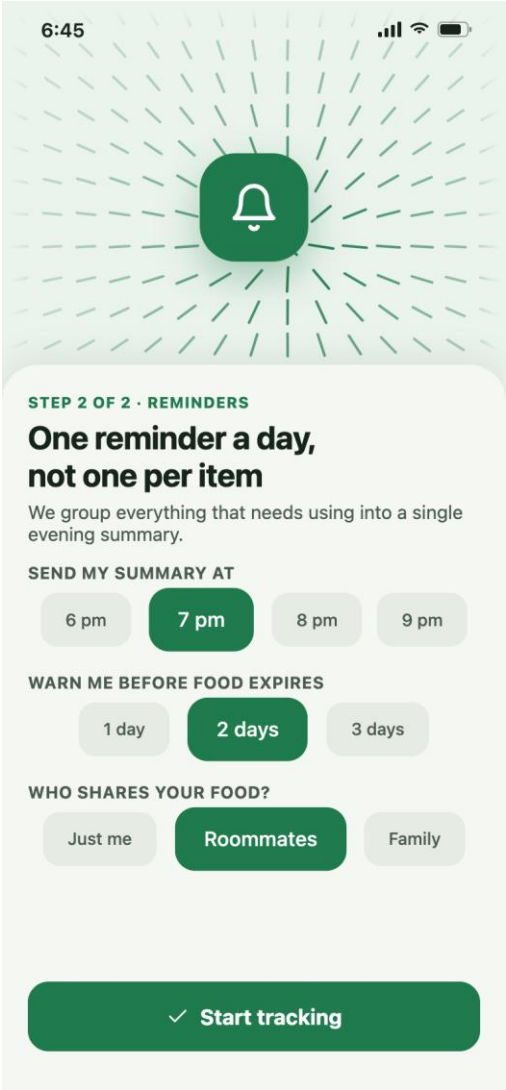


Figure 9. S2 Reminder setup

Welcome introduces the purpose in one sentence. Get started leads to a single setup step; returning users can skip it. Reminder setup asks only the three questions that change behaviour later, using large chip controls, and explains why there is only one reminder a day.

Final screens 3 and 4

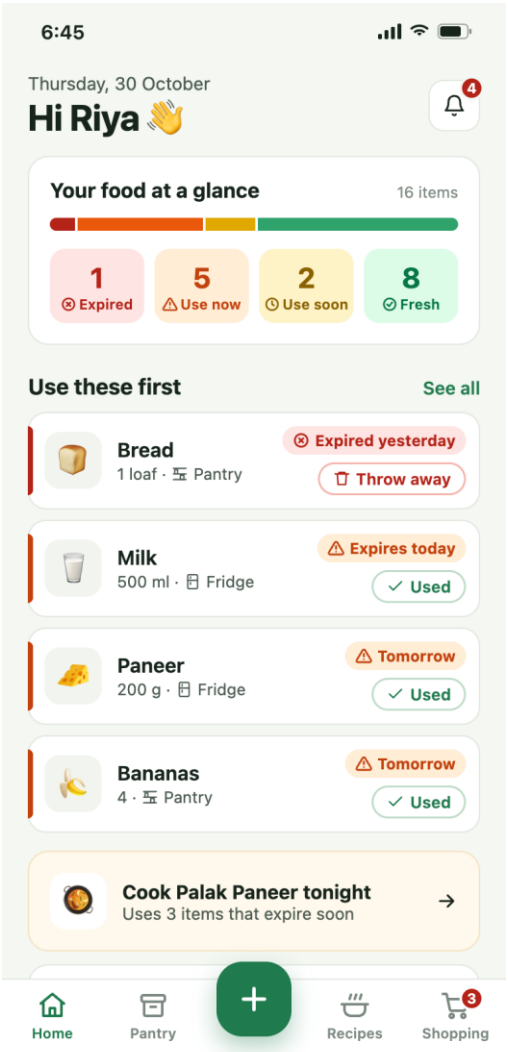


Figure 10. S3 Home

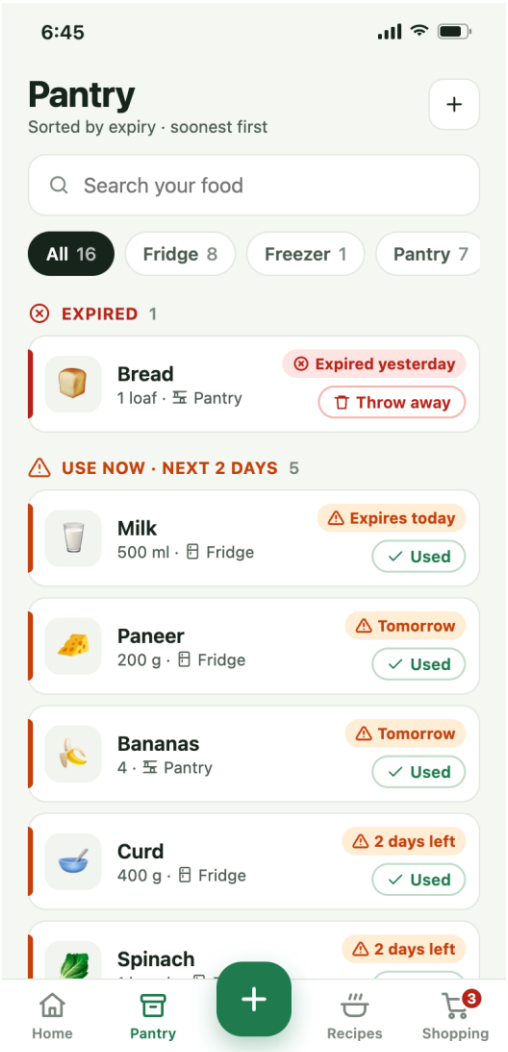


Figure 11. S4 Pantry inventory

Home answers “what should I use first?”: a summary bar and four counts (each opens the filtered pantry), the four most urgent items with a one-tap Used action, and a recipe that uses expiring food. The pantry groups items by urgency, soonest first, with location filters.

Final screens 5 and 6

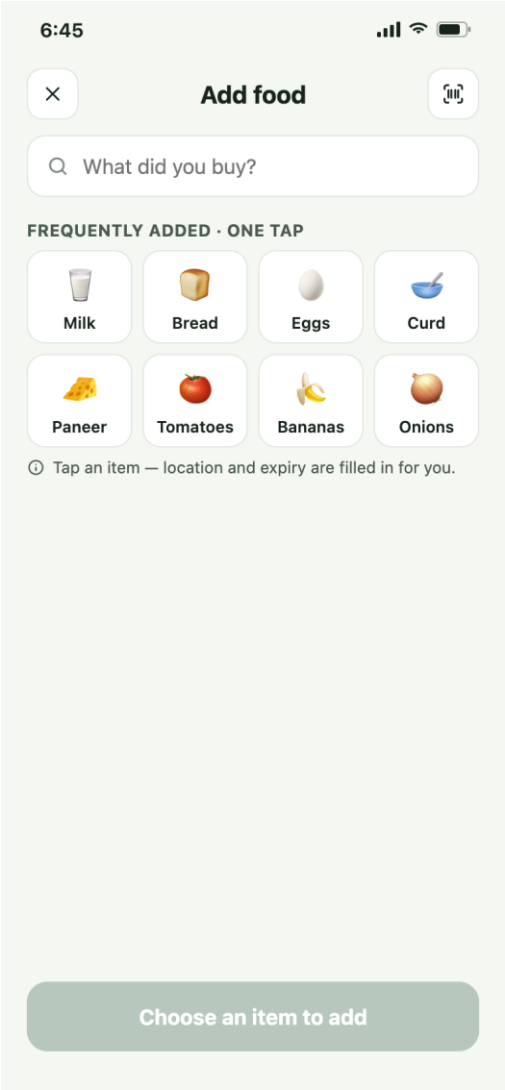


Figure 12. S5 Add food (quick add)

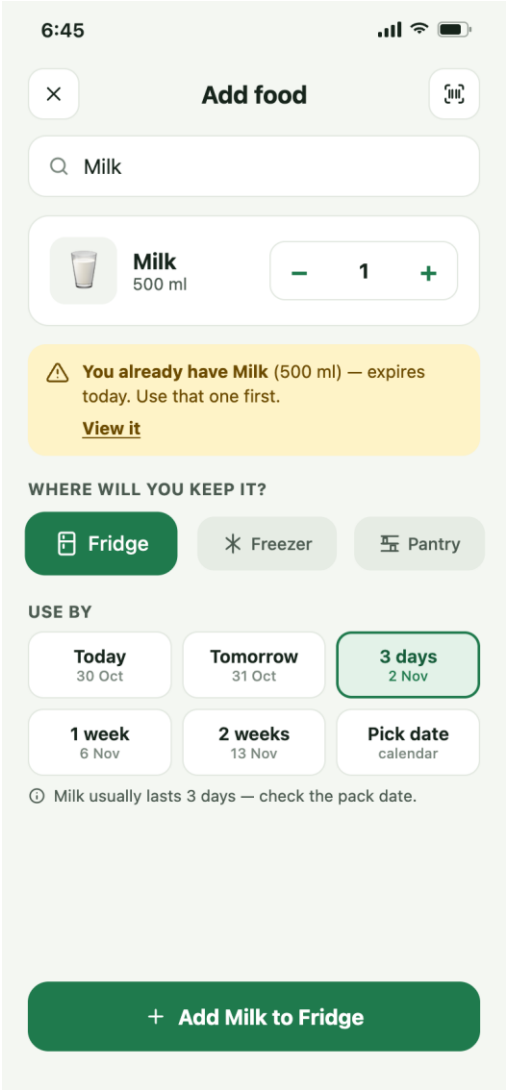


Figure 13. S6 Add — duplicate warning

Quick add shows eight frequent items; one tap fills in location and use-by date, with a hint about typical shelf life. When milk is already in the fridge, an amber warning appears before saving, but adding is still allowed.

Final screens 7 and 8

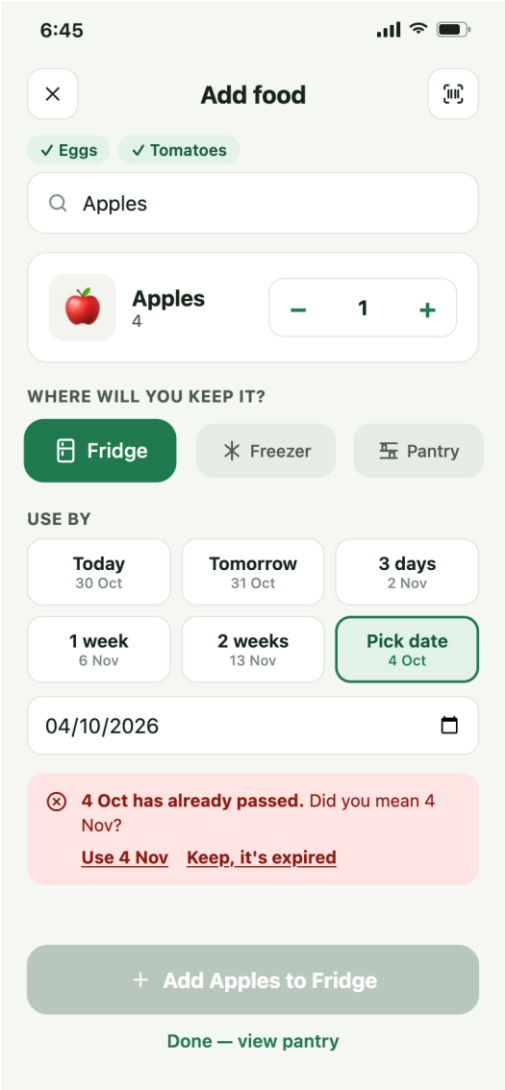


Figure 14. S7 Add — past date prevented

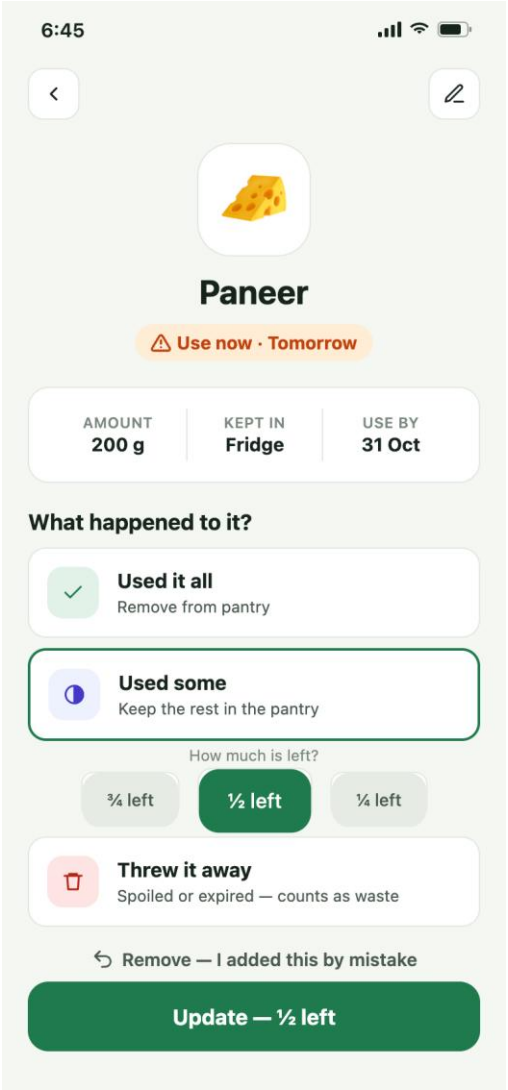


Figure 15. S8 Item — mark as used

A past date (4 Oct, probably a typo for 4 Nov) is caught and a one-tap fix is offered. Items added in this session are shown as chips at the top. The item screen asks “What happened to it?” and separates used, partly used and wasted food.

Final screens 9 and 10

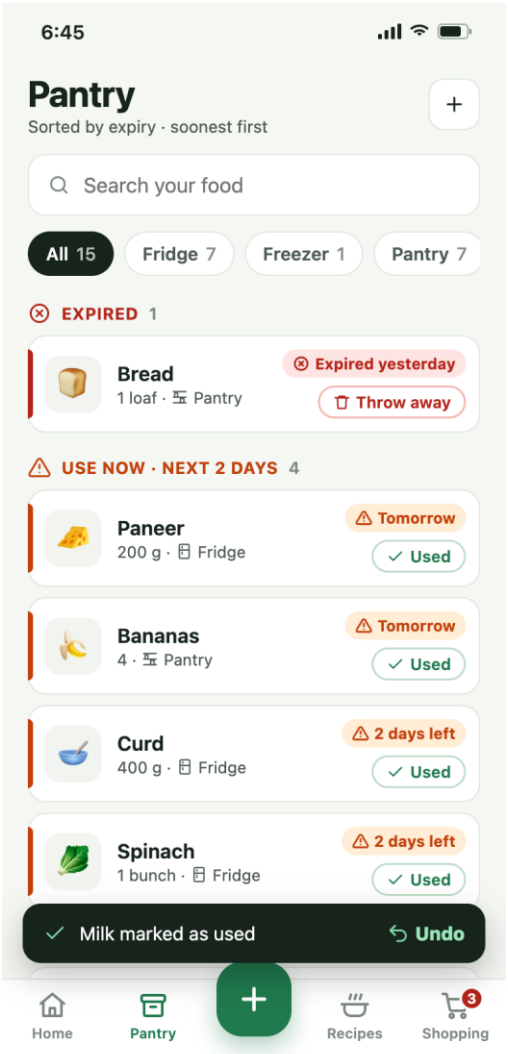


Figure 16. S9 Marked as used, with Undo

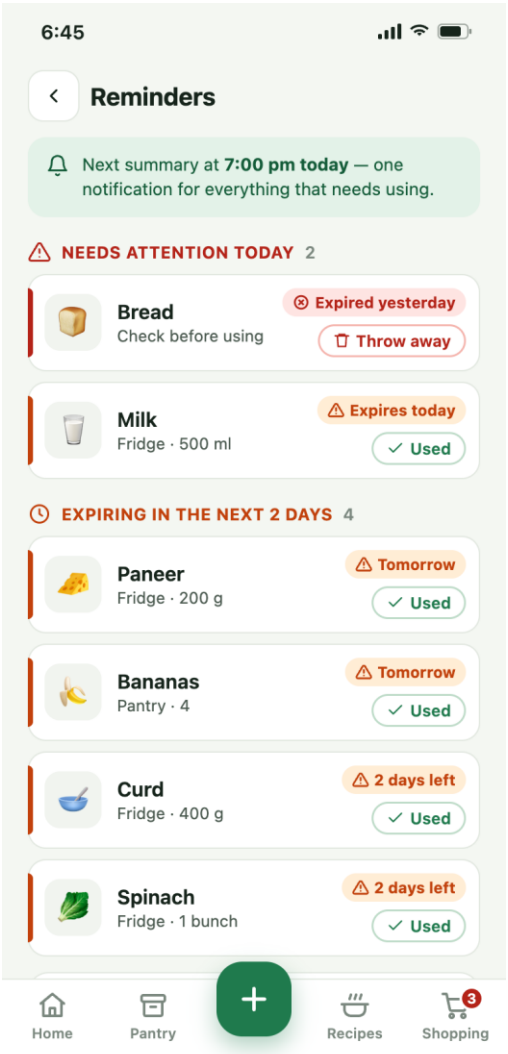


Figure 17. S10 Expiry reminders

After marking milk as used, the item disappears and a snackbar offers Undo for 5 seconds instead of a confirmation dialog. Reminders group everything that needs attention today and in the chosen warning period, with the settings on the same page.

Final screens 11 and 12

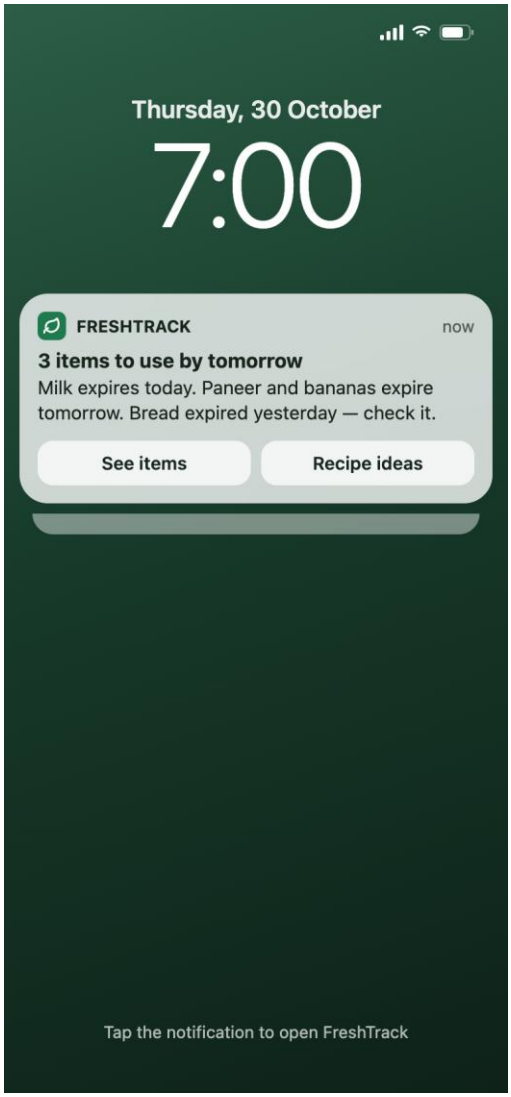


Figure 18. S11 Lock-screen summary

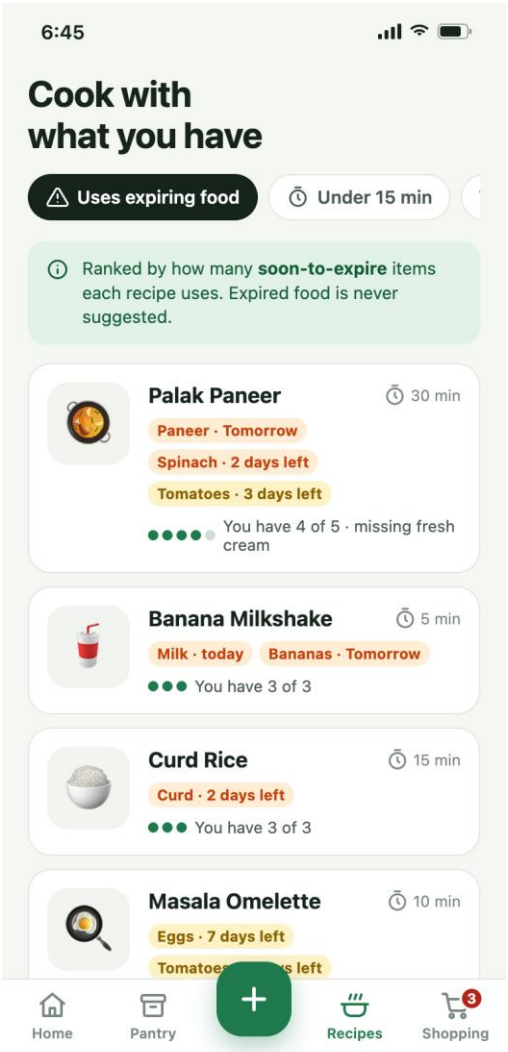


Figure 19. S12 Recipe suggestions

At 7 pm one grouped notification lists the items that need using, instead of one alert per item. Recipes are ranked by how many soon-to-expire items they use, and each card shows which items those are and how many ingredients are at home.

Final screens 13 and 14

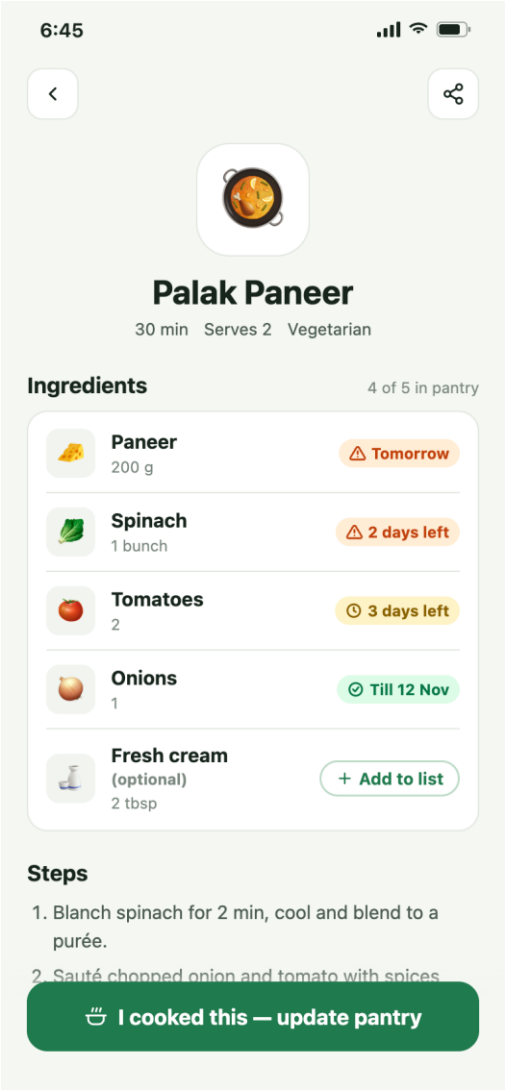


Figure 20. S13 Recipe detail

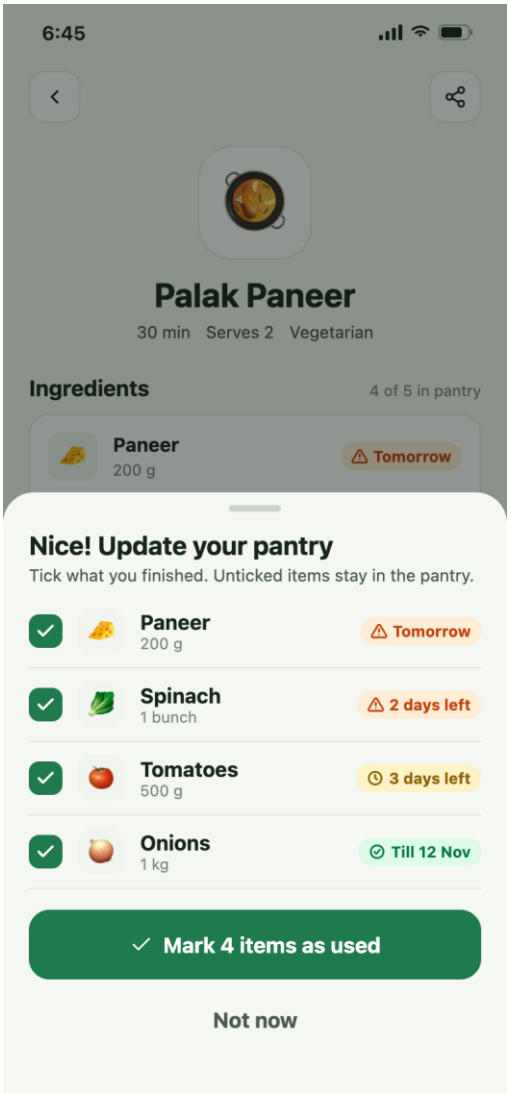


Figure 21. S14 Cooked — update pantry

The recipe shows ingredients with their urgency and adds missing ones to the shopping list in one tap. “I cooked this” opens a sheet where the used ingredients are already ticked, so the pantry is updated in one step.

Final screens 15 and 16

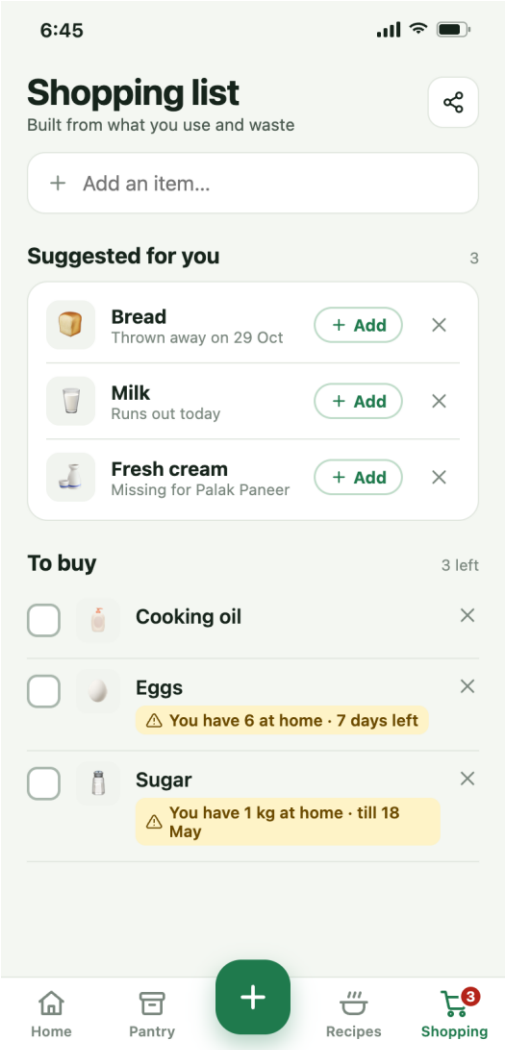


Figure 22. S15 Shopping list

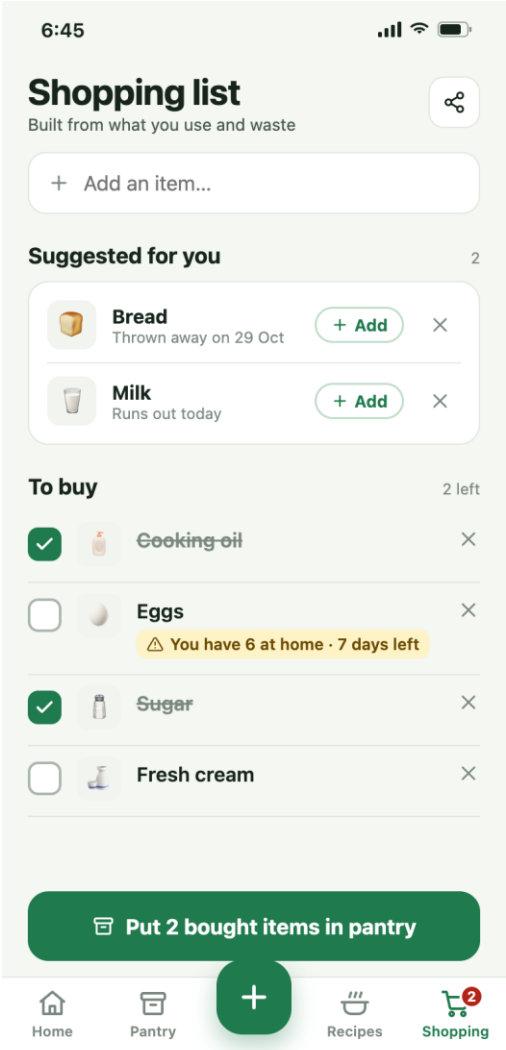


Figure 23. S16 Bought → pantry

The shopping list suggests items with a reason (thrown away, runs out, missing for a recipe) and warns when something on the list is already at home. Ticked items go to the pantry in one tap, with usual expiry dates filled in.

6. Human-Centred Design Principles Applied

6.1 Understand Users

The design started from the user's goal (eat food before it spoils, with little effort) rather than from the data an inventory system needs. The context-of-use analysis, HTA and personas in Sections 2–3 determined the priorities: adding must be faster than the effort it saves, urgency must be readable in a glance, and reminders must respect attention. Human memory is limited, so FreshTrack relies on recognition rather than recall: frequent items are shown instead of typed, and the app remembers dates so the user does not have to.

6.2 User Involvement

Users were not directly involved in this design cycle, which is a limitation. The personas and empathy map stood in for users when choosing defaults, wording and reminder behaviour. The interactive prototype is ready for the planned usability test in Section 7.2, which is the next point at which real users will shape the design.

6.3 Iterative Design

The design went through three iterations: sketches (Figure 5) fixed the concept, the v1 wireframes (Figure 6) made the structure concrete, and inspection of v1 with heuristic evaluation, cognitive walkthrough and KLM produced the final version. The changes are traceable from the numbered issues in Figure 6 to Table 15 and the final screens. The interactive prototype was also adjusted while it was being built, for example by shortening chip labels that did not fit on small screens.

6.4 User Evaluation

Evaluation took place throughout the design, not only at the end. Expert evaluation (Nielsen's heuristics, cognitive walkthrough and GOMS/KLM prediction [2], [5], [6]) was applied to the v1 wireframes and repeated on the final design. Section 7 records the findings, and a usability specification sets the measurable targets for user testing.

6.5 Whole User Experience

The design covers the whole experience around stored food, not only the inventory screen: the moment of putting groceries away (quick add), the daily decision about what to eat (reminder and recipes), the physical action of cooking or throwing food away (mark as used / wasted), and the next shopping trip (generated list and restocking). Eaten and wasted food are recorded separately, so the monthly feedback on Home reflects what actually happened. The one-time setup and the tone of the wording ("Nice! Update your pantry") aim to make the habit pleasant rather than a chore.

Selected design rules from the course were applied explicitly:

Table 10. Design rules applied

Design rule	Where it is applied
Shneiderman — strive for consistency	Same four priority levels, badges and words on Home, Pantry, Reminders, Recipes and in the notification.
Shneiderman — offer informative feedback	Toasts after every action ("Milk marked as used"); chip swell on selection; counts update immediately.
Shneiderman — design dialogs to yield closure	"Added this session" chips and "Done — view pantry"; "Put 2 bought items in pantry".
Shneiderman — prevent errors; permit easy reversal	Duplicate and past-date checks; Undo for 5 s after every removal instead of a confirmation dialog.
Shneiderman — reduce short-term memory load	Frequent-item grid and autocomplete; relative dates ("2 days left") instead of raw dates.
Norman — make things visible; bridge the gulfs	Action labels describe the result ("Add Milk to Fridge"); the badge shows state without opening the item.
Norman — knowledge in the world	Typical shelf life shown as a hint; reasons shown on shopping suggestions.
Dix — learnability, flexibility, robustness	Familiar words (predictability); tap, search or scan to add (substitutivity); undo and warnings (recoverability).

7. Design Evaluation

7.1 Heuristic Evaluation

The final screens were inspected against Nielsen's ten usability heuristics [5]. Severity uses Nielsen's 0–4 scale for problems that remain (0 = none, 4 = catastrophe).

Table 11. Heuristic evaluation of the final design

Heuristic	Observation on the final design	Sev.
Visibility of system status	Priority badge on every item; counts on Home; next summary time shown on Reminders; toast after each action.	0
Match with the real world	Fridge / Freezer / Pantry; "Used it all", "Threw it away"; relative days instead of dd/mm/yy.	0
User control and freedom	Undo for 5 s after any removal; "Remove — I added this by mistake"; skip on Welcome.	1
Consistency and standards	One badge system everywhere; primary action always bottom-full-width; standard tab bar.	0
Error prevention	Duplicate warning, past-date check, "already at home" warning, expired food excluded from recipes.	0
Recognition rather than recall	Frequent-item grid, autocomplete, ingredients shown with their pantry status.	0
Flexibility and efficiency	Tap, search or scan to add; quick "Used" on rows for experts, full detail screen for partial use.	1
Aesthetic and minimalist design	Colour is reserved for urgency; decorative motion only on first-run screens.	1
Help users recover from errors	Errors state the problem and offer a fix ("Did you mean 4 Nov? Use 4 Nov").	0
Help and documentation	Inline hints (typical shelf life, how ranking works); no separate help section yet.	2

Cognitive walkthrough

We walked through task T1 for Persona A (add milk, eggs and tomatoes after shopping) and task T3 (mark the paneer as used) on the final prototype, asking the four walkthrough questions [6] at every step: (Q1) will the user try to achieve the right effect, (Q2) will they notice the correct action is available, (Q3) will they associate the action with the effect, and (Q4) will they see progress after the action?

Table 12. Cognitive walkthrough (✓ = yes, ~ = partly)

Step	Q1	Q2	Q3	Q4	Note
T1.1 Tap (+)	✓	✓	✓	✓	Central, largest button in the tab bar.
T1.2 Tap "Milk"	✓	✓	✓	✓	Item card and defaults appear; duplicate warning seen.
T1.3 Accept defaults	✓	✓	~	✓	User may not know defaults are estimates → hint "usually lasts 3 days — check the pack".
T1.4 Tap "Add Milk to Fridge"	✓	✓	✓	✓	Label states the result; toast and "✓ Milk" chip.
T1.5 Repeat for eggs, tomatoes	✓	✓	✓	✓	Form resets, grid visible again.
T3.1 Find paneer on Home	✓	✓	✓	✓	Listed under "Use these first" with "Tomorrow".
T3.2 Tap "Used"	✓	✓	~	✓	Quick action removes all of it; partial use needs the detail screen → Undo covers mistakes.

Keystroke-level model (KLM) comparison

Using the KLM operators from the course [2] (M = 1.35 s, P = 1.1 s per tap, K = 0.28 s per key), we predicted expert times for the inventory workflow in the v1 wireframes and in the final design.

Table 13. KLM predictions

Task	v1 wireframe	Final design	Change
Add "Milk" (1 item)	6M + 10P + 13K = 22.7 s (6 fields, typed date)	2M + 2P = 4.9 s (tap item, check, add)	−78%
Add 10 items after shopping	≈ 227 s	≈ 49–60 s	−74% or more
Mark an item as used	2M + 3P = 6.0 s (open, Delete, confirm)	M + P = 2.5 s (tap "Used" on the row)	−59%
Find the most urgent item	Scan an A–Z list and compare dates	Top of Home, labelled	No date arithmetic

7.2 Usability Testing

Usability testing with participants has not yet been conducted. We prepared a test plan and a usability specification (Table 14) so that the prototype can be tested with five to

eight students and household members, using the think-aloud protocol followed by the System Usability Scale (SUS) questionnaire [7].

- **Task A:** You have just bought milk, eggs and tomatoes. Add them to FreshTrack.
- **Task B:** Without opening any item, tell us which three foods you should use first.
- **Task C:** You drank the milk. Update the app. Then undo it — you actually drank only half.
- **Task D:** Find something to cook tonight that uses food about to expire, and update the pantry after cooking.
- **Task E:** Prepare for shopping. Is there anything on the list you do not need to buy?

Table 14. Usability specification for the test

Attribute	Measuring method	Now level (v1)	Worst	Planned	Best
Efficiency — add an item	Time per item, Task A	≈ 23 s (KLM)	10 s	6 s	4.9 s
Effectiveness — prioritisation	Correct "top 3" in Task B within 10 s	Not supported	70%	90%	100%
Error prevention	Wrong dates or duplicates saved in Task A	Not prevented	1	0	0
Recoverability	Actions needed to undo a wrong removal	Impossible	3	1	1
Satisfaction	SUS score (0–100)	—	68	75	85

7.3 Issues Identified

Inspection of the v1 wireframes (Figure 6) identified the following issues. The numbers match the markers in Figure 6.

Table 15. Issues identified in the initial design

#	Issue in initial design	Potential effect	Response in final design
1	Inventory sorted A–Z	Urgent items hidden mid-list; milk found too late	Sort by expiry; group into four priority levels (S4)
2	Expiry shown only as dd/mm/yy	User must work out days left; easy to misread	Relative wording plus colour and icon badge (Figure 7)
3	Six required fields per item	≈ 23 s per item; users stop recording food	One-tap frequent items with defaults; search; scan (S5)
4	Free date entry, no checks	Past or mistyped dates saved silently	Use-by presets; past-date detection with fix (S7)
5	Only "Delete" + confirmation	Eaten and wasted food indistinguishable; dialog habitually confirmed	Used / Used some / Threw away / Mistake; Undo (S8, S9)
6	One notification per item	Alert fatigue; all notifications dismissed	One grouped summary at a chosen time (S2, S11)
7	Manual shopping list	Items at home bought again	Generated suggestions; "already at home" warning (S15)

Inspection of the final design also found smaller issues that remain (severity 1–2 in Table 11): there is no separate help section; the quick "Used" button removes the whole item, so partial use needs the detail screen; and recipe filter chips other than the default are not yet functional in the prototype.

8. Design Improvements

The final version addresses every issue in Table 15 without adding steps to the main journeys. The most important changes are:

- **Visual prioritisation.** The inventory is sorted by expiry and grouped into four levels, each shown with colour, icon and words. Home turns this into a glanceable summary (S3, S4, Figure 7).
- **Minimal data entry.** Adding an item went from six fields to one tap plus confirmation, with defaults from typical shelf life and a session list for adding many items in a row (S5). Predicted time per item fell from about 23 s to about 5 s.
- **Mistake prevention.** Duplicate, past-date and "already at home" checks warn before the mistake happens. They are warnings, not blocks, so the user stays in control (S6, S7, S15).
- **Reversible actions.** The Delete-and-confirm pattern was replaced by meaningful outcomes (used, used some, thrown away, added by mistake) and a 5-second Undo (S8, S9).
- **Reminders that respect attention.** One grouped evening summary at a chosen time, with a 1–3-day warning period set once during setup (S2, S10, S11).
- **Closing the loop.** Recipes use expiring food and update the pantry in one step; the shopping list is generated from what was used, wasted or missing, and bought items return to the pantry with dates filled in (S12–S16).

Remaining limitations: real notification delivery, barcode lookup and recipe data are simulated; sharing between roommates is represented by a share button only; and the design has not yet been tested with screen readers or with real users.

9. Final Prototype

The final prototype consists of sixteen mobile screens on a 390 × 844 logical-pixel canvas (Figures 8–23). It is also built as a working, clickable prototype (one self-contained HTML file) in which the full journey can be tried: items can be added, marked as used, undone, cooked and restocked, and every count and list updates. The interactive chip controls and the first-run animations use four open-source React Bits components: JellyRadio, InfiniteSpiral, MagicRings and MagnetLines [8].

Table 16. Final screens and requirements covered

Screen	Purpose	Requirements
S1–S2	First run: welcome and one-time reminder setup	FR7
S3–S4	Home summary and priority-sorted pantry	FR4, FR5
S5–S7	Quick add with defaults, duplicate warning and past-date prevention	FR1–FR3
S8–S9	Mark as used / used some / thrown away; undo	FR6
S10–S11	Grouped reminders and lock-screen summary	FR7
S12–S14	Recipe suggestions, recipe detail, update after cooking	FR8, FR9
S15–S16	Generated shopping list and restocking	FR10

High-resolution PNG exports (2×) of all sixteen screens and of the sketches, wireframes and diagrams are supplied with this report.

10. Conclusion

FreshTrack demonstrates how a food inventory system can be designed around the user's actual needs, focusing on the question, "What should I use first?" rather than simply collecting and managing data. By following the ISO 9241-210 human-centred design approach, we analysed the context of use, identified user requirements, and translated them into practical design solutions. The development process progressed systematically from initial sketches to wireframes and finally to a high-fidelity interactive prototype. The final design prioritises clarity, efficiency, and ease of interaction. It makes food urgency visible at a glance, reduces the time required to record an item by approximately three quarters, minimises common user errors, and ensures that every removal can be reversed.

Overall, the project highlights the importance of understanding user behaviour and addressing everyday challenges through thoughtful interface design. It also demonstrates how usability principles can transform a routine household activity into a simpler, more organised, and efficient process. The emphasis throughout has remained on reducing unnecessary effort while helping users make informed decisions about their food supplies.

The next step is to conduct the usability testing planned in Section 7.2 with students and household members, followed by further refinements based on their feedback and observed interactions. Future iterations could introduce shared pantries for roommates, integration with real-world barcode databases and recipe platforms, and improved accessibility through screen-reader support. These enhancements would help FreshTrack evolve into a more inclusive, collaborative, and practical food management solution.

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